Xiao Liu

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RESEARCH SUMMARY

Computer Science PhD student focused on generative AI, diffusion and flow-based models, reinforcement learning, and secure large language models. I am passionate about the mathematical and algorithmic foundations of AI and their applications to scientific discovery and trustworthy AI systems. My research combines principles generative modeling with predictive signals and RL to solve real-world problems, such as protein/antibody design and model security.

EDUCATION

Shanghai Jiao Tong University — *Shanghai, China*
PhD in Computer Science and Technology — *Apr. 2025 – Present*

University College London & University of Cambridge — *London & Cambridge, UK*
MRes in Connected Electronic and Photonic Systems, Merit — *Sep. 2020 – Sep. 2021*

University of Nottingham Ningbo China (UNNC) — *Zhejiang, China & Nottingham, UK*
Beng in Electrical and Electronic Engineering, First Class Honours — *Sep. 2016 -- May. 2020*

RESEARCH EXPERIENCE

GABIF: An Adaptive Guided Graph Denoising Framework for Antibody Conditional Inverse Folding — **2025 - Present**

NeurIPS 2026 *submission; joint first author, second-listed author*

- Co-designed a guided discrete diffusion framework for conditional sequence generation and property-aware optimization.
- Co-led the algorithmic design, experimental protocol, benchmark evaluation, ablation studies, and testing pipeline.
- Integrated external predictive signals into the generative sampling process to improve downstream target properties.
- Co-wrote and revised the manuscript with the other joint first author.

Permutation-based Transformer Model Protection — **2025 - Present**

Ongoing exploratory research on secure and privacy-preserving model architectures

- Exploring permutation-equivalence mechanisms to enhance Transformer architecture confidentiality under black-box and white-box attacks.
- Investigating model transformation strategies that preserve functional behavior while increasing resistance to model extraction.

Signal Processing for Cuff-less Blood Pressure Measurement Using PPG — **May 2020 – Aug. 2021**

Supervisors: Prof. Andreas Demosthenous; Prof. Dai Jiang

- Proposed a machine learning framework to improve cuff-less blood pressure estimation from PPG signals.
- Implemented PPG signal prediction and restoration with CNN/LSTM-based models in TensorFlow.
- Modeled nonlinear relationships between PPG features and blood pressure using deep learning methods.

WORK EXPERIENCE

Huawei Technologies Co., Ltd. — *Engineer, Machine Learning* — *June. 2022 – Oct. 2024*

- Designed and implemented machine learning algorithms to improve GNSS positioning accuracy under complex terminal scenarios.
- Led large-scale C software project refactoring to improve code modularity, integrity, maintainability, and system reliability.
- Provided algorithm optimization and technical support for terminal products including smartphones and smart watches

TECHNICAL SKILLS

Programming	Python, C/C++, Bash, LaTex
Machine Learning	PyTorch, TensorFlow/Keras, Scikit-learn, Transformers
Generative AI	Diffusion Models, Flow-based Models, Graph Neural Networks, Protein/Scientific Generative Modeling
LLM Systems	vLLM, Model Serving, Local LLM Deployment
Tools & Platforms	Linux, Docker, Git, VS Code, Conda

AWARDS & HONORS

- Best QCC Award (2nd Prize), Huawei Technologies
- Xinxing Award (twice), HiSilicon
- Nottingham Advanced Award (UNNC)
- UNNC Dream Scholarship (2019)